



Arrays, 2D & 3D, and ArrayLists

Intermediate Object-Oriented Programming · Java

Core Java, Vol. I & II (13th ed.) · three days, ~3.75 hours · most of it spent running code together

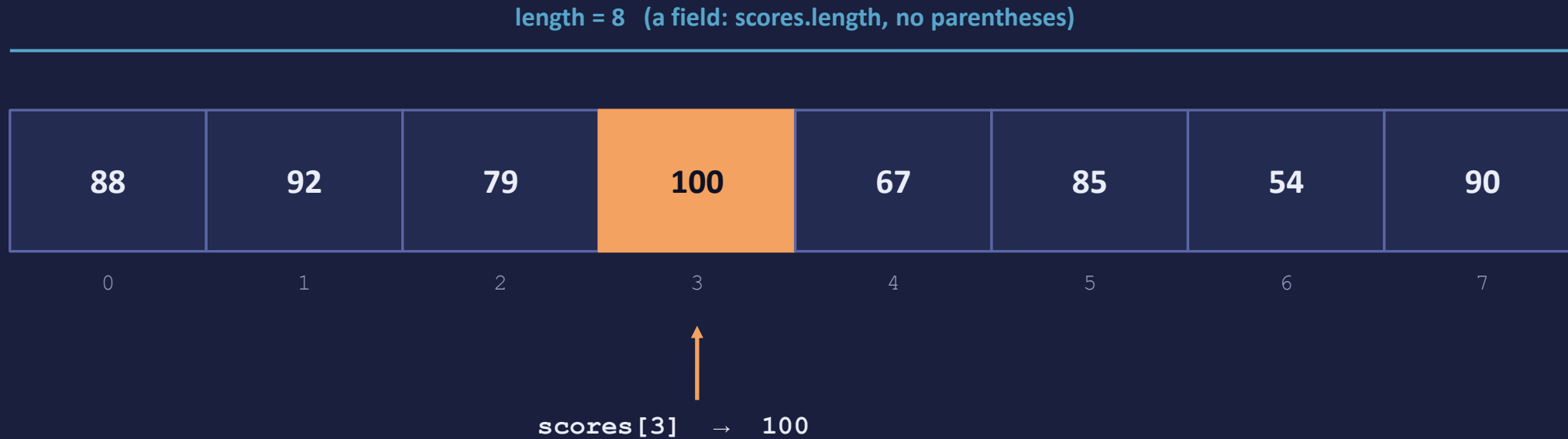


The arc



The slides give you the picture; class time is for predicting what code prints, then running it to find out.

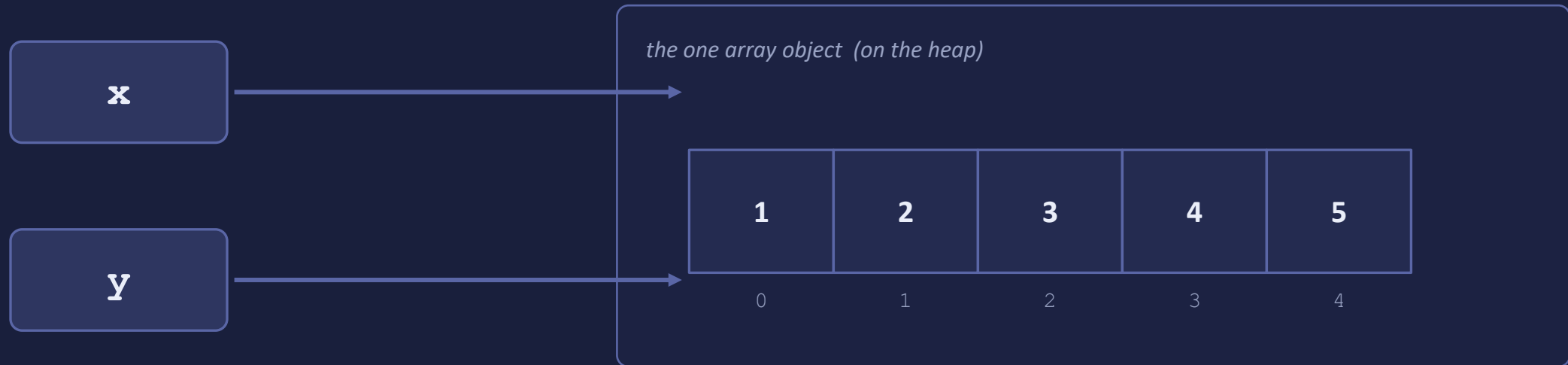
An array is a row of indexed boxes



```
int[] scores = new int[8];    // 8 slots, each defaults to 0
```

Indices run 0 ... length-1. The last valid index is length-1 — never length.

The variable is a label, not the box



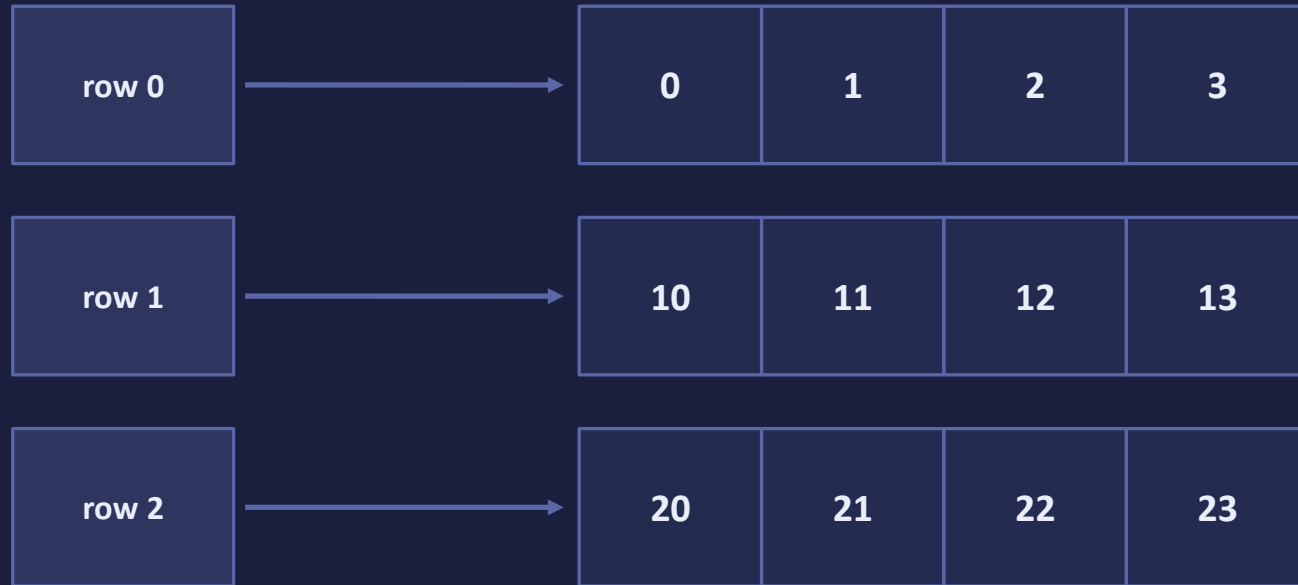
```
int[] y = x;    // y and x now point at the SAME array
```

Change through y and x sees it too. Want an independent copy? `Arrays.copyOf(x, x.length)`.

A 2-D array is an array of arrays

grid

grid[0].length = 4



grid.length = 3

```
int[][] grid = new int[3][4];    // 3 rows, each its own int[4]
```

Internalize this: the outer array holds references to rows. Everything else follows.

Indexing: `grid[row][col]`

		col →			
		0	1	2	3
row ↓	0	0,0	0,1	0,2	0,3
	1	1,0	1,1	1,2	1,3
	2	2,0	2,1	2,2	2,3

← `grid[1][2]`

*first bracket = which row,
second = which column*

grid.length is the number of rows; grid[r].length is that row's width.

Rows need not be the same length

`tri`



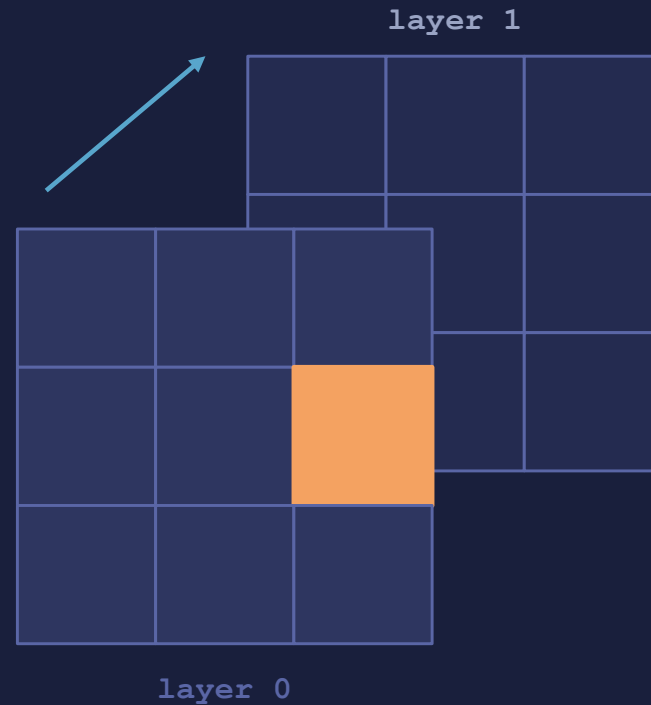
```
int[][] tri = new int[4][]; // 4 row  
slots — each row still null
```

Then allocate each row on its own:
`tri[r] = new int[r + 1];`

Natural fit for a triangle, a lower-triangular matrix, or "games each player played this week."

Right after `new int[4][]`, every row is null — the outer array exists, the rows don't yet.

Stack grids into layers



Read the indices outside-in:

`cube[ layer ]`

which grid

`cube[ layer ][ row ]`

which row in it

`cube[ layer ][ row ][ col ]`

which column

```
int[][][] cube = new int[2][3][4];
```

Rare but real: voxel worlds, stacks of image frames, values sampled over time.

The Arrays class does the busywork



`Arrays.sort(a)`

orders it in place, ascending



`Arrays.fill(a, v)`

sets every slot to v



`Arrays.copyOf(a, n)`

"resize" — new array, copied over



`Arrays.equals(a, b)`

same contents? (not same object)



`Arrays.binarySearch(a, k)`

fast lookup — sorted arrays only



`Arrays.deepToString(a)`

readable printing of nested arrays

Stop hand-coding this. Learn these six and you'll rarely write array bookkeeping again.

When the count won't hold still

array



length frozen at creation — no more, no fewer

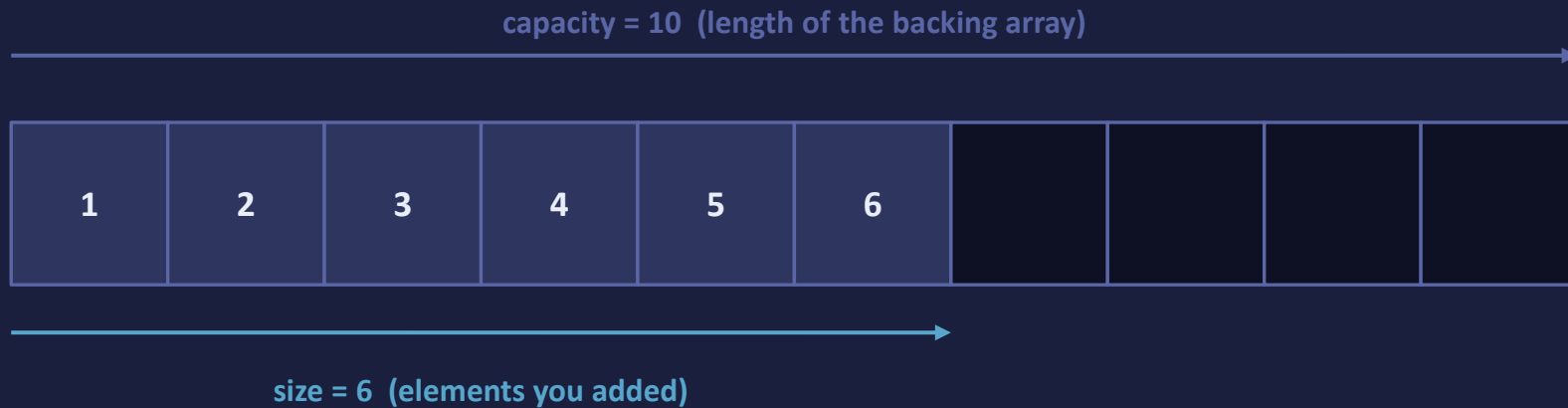
ArrayList



add() when you need more; remove() when you don't

An ArrayList wraps an array and, when it fills up, quietly copies into a bigger one for you.

Size is what you added; capacity is the room



add past capacity → allocate $\approx 1.5\times$ and copy (grow by half, NOT double)



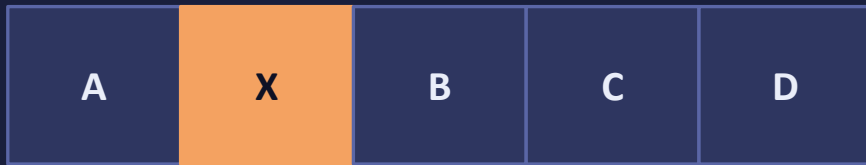
Default capacity 10, allocated on the first add. Know the count up front? `new ArrayList<>(n)` skips the copies.

Insert and remove shift the neighbors

```
list.add(1, "X")
```



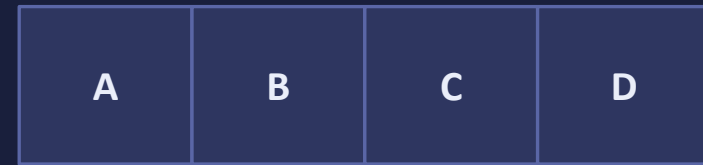
B, C, D slide right →



```
list.remove(1)
```

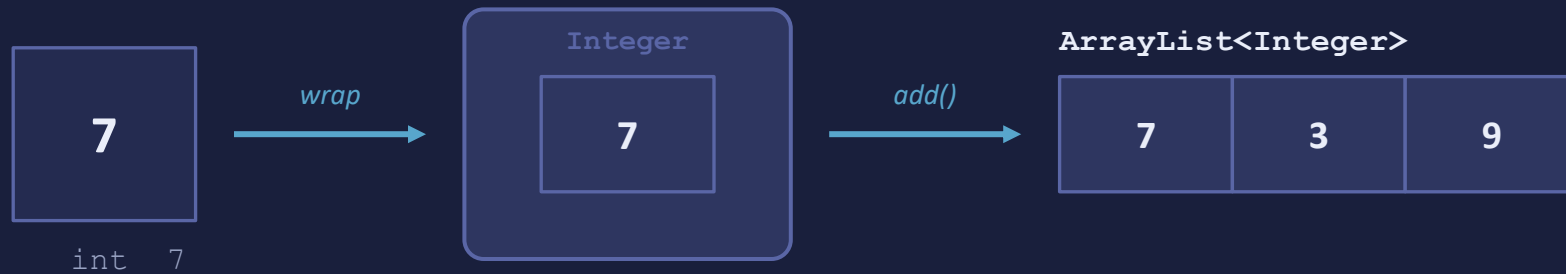


← B, C, D slide left



Appending at the end is cheap. Inserting or removing in the middle moves everything after it.

There is no `ArrayList<int>` — Java boxes for you



```
ArrayList<Integer> nums = new ArrayList<>();  nums.add(7);  int x = nums.get(0);
```

Numbers auto-wrap going in and auto-unwrap coming out. Convenient — but each becomes a heap object.

Millions of numbers in a hot loop? A plain `int[]` is the honest choice.

remove(2) is not remove(Integer.valueOf(2))



On an ArrayList<Integer>, an int argument calls remove(int index); an Integer calls remove(Object). Memorize this one cold.

Array or ArrayList?

Reach for an array

- size is fixed and known
- primitives in a performance-sensitive spot
- grids, boards, matrices (multi-dimensional)
- an API hands you or wants arrays

Reach for an ArrayList

- the count grows or shrinks at runtime
- you don't know the count up front
- you want contains / indexOf / remove
- the everyday "a bunch of things" need

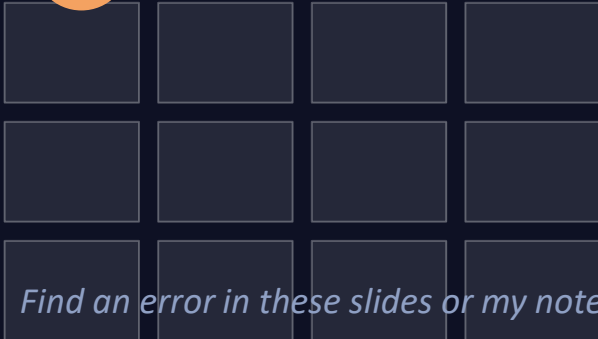
Rule of thumb: fixed grid of primitives → array. Growing pile of objects → ArrayList.

Your turn

1 **Read** the arrays & ArrayList material in the book — and my notes

2 **Practice** run every example, then break it and see what happens

3 **Demo** build a project and show it to me working



Find an error in these slides or my notes? Tell me — that's the assignment behind the assignment.